DHAVALA SAI MADHAV

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CAREER OBJECTIVE

AI/ML Engineer & Full Stack Developer skilled in Machine Learning, Generative AI, RAG, and modern web development. Interested in building scalable AI-driven solutions for real-world problems.

EDUCATION

Vignan's Institute of Information Technology (VIIT) *2023 – 2027*
B.Tech in Computer Science and Engineering **CGPA: 9.27**

EXPERIENCE

AI/ML Intern — SmartBridge Educational Services Pvt. Ltd. *Aug 2026*
Python, Machine Learning, Scikit-learn, FastAPI, Pandas, NumPy

- Completed a 2-month (120-hour) AI/ML internship organized by SmartBridge Educational Services Pvt. Ltd. in collaboration with APSCHE, developing **OptiCrop**, a machine learning-based crop recommendation application.
- Worked on data preprocessing, model training, evaluation, and FastAPI-based deployment to integrate ML predictions into an interactive web application.

AI/ML Full Stack Intern — HalfTone Systems Startup *Feb 2026 – May 2026*
React.js, Node.js, Express.js, JavaScript, MongoDB, Git, GitHub, Vercel

- Developed and maintained full-stack web applications and AI-driven features, including the company's official website and VayuCare Forge healthcare platform.
- Collaborated on development, testing, and deployment of production-ready applications using modern frontend, backend, and version-control workflows.

FREELANCER — Vexamo *Remote — Aug 2025 – Present*
Full Stack Development, React.js, Node.js, Express.js, MongoDB

- Built 5+ responsive full-stack web applications and interactive dashboards, improving workflow efficiency and client engagement for professionals and educational institutions.

PROJECTS

Toleran — Engineering Drawing Intelligence & Design Compliance Copilot [GitHub]
Python, FastAPI, OpenCV, PyMuPDF, OCR, LangChain, RAG, PostgreSQL, LLMs

- Built an AI-powered platform to extract dimensions and tolerances from engineering drawings and technical specifications using document processing, OCR, and structured analysis.
- Developed rule-based compliance checks and a RAG-powered engineering copilot to identify discrepancies, explain findings with source evidence, and support human review.

AI Conferencing — Meeting Intelligence Platform [Live] [GitHub]
React.js, Node.js, Express.js, FastAPI, MongoDB, JWT, ChromaDB, LangChain

- Built an AI-powered conferencing platform with real-time speech transcription, semantic search, and meeting-specific chatbots for recorded meetings.
- Implemented a RAG architecture using ChromaDB and LangChain to summarize transcripts and answer meeting-specific questions.

Advanced RAG — Retrieval-Augmented Generation System [GitHub]
Python, LangChain, FastAPI, ChromaDB, Hugging Face, LLMs

- Developed a Retrieval-Augmented Generation (RAG) system combining document retrieval, vector embeddings, and large language models for context-aware responses.
- Implemented semantic search using sentence-transformer embeddings and ChromaDB, with FastAPI for document ingestion, retrieval, and LLM-powered response generation.

TECHNICAL SKILLS

Programming: Python, JavaScript, C
AI/ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, Transformers, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)
Computer Vision & Document AI: YOLO, OpenCV, Image Processing, Object Detection, OCR, PyMuPDF
Frameworks & Libraries: Scikit-learn, TensorFlow, LangChain, Nginx
Web Technologies: Node.js, Express.js, React.js, FastAPI, MongoDB
Tools & Platforms: Git, GitHub, Streamlit, Google Colab, VS Code, Docker, Vercel, Render
Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP, System Design

CERTIFICATIONS & ACHIEVEMENTS [Certificates]

- **Cisco Networking Academy** — Python Essentials, Full Stack (HTML, CSS, JavaScript), CCNS.
- Achieved a **2-Star Rating on CodeChef** and solved **160+ coding problems** across LeetCode, CodeChef programming platforms.